

Build the lab. Run the tests. Verify the logs.

Step-by-step instructions with tables and real screenshots

Windows 11 Enterprise Evaluation | Hyper-V | Event Viewer

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Manual log analysis | Three controlled authentication tests

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Create the virtual machine

Start on the physical PC. The original host used Windows 11 Pro and 32 GB RAM.

- 1 Check virtualisation.** Open **Task Manager > Performance > CPU**. Confirm **Virtualization: Enabled**. If disabled, enable hardware virtualisation in the PC firmware before continuing.
- 2 Enable Hyper-V.** Open **Turn Windows features on or off**, select **Hyper-V** with its management tools and platform, then restart. This guide assumes Windows 11 Pro or another edition supporting Hyper-V.
- 3 Download the guest OS.** Get the Windows 11 Enterprise Evaluation **x64 ISO in English** from the [Microsoft Evaluation Center](#). Keep the ISO available on the host.
- 4 Run the VM wizard.** Open **Hyper-V Manager**, select your host, then **New > Virtual Machine**. Use the settings below and click Finish.

Wizard page	Value to enter
Specify Name and Location	SOC-Lab-Windows; choose a suitable storage folder
Specify Generation	Generation 2
Assign Memory	8192 MB; leave Dynamic Memory unchecked for this setup
Configure Networking	Default Switch
Connect Virtual Hard Disk	Create a VHDX; 80 GB, dynamically expanding
Installation Options	Install from a bootable image file; select the downloaded ISO

- 5 Set CPU and security.** With the VM off, open **Settings > Processor** and set 2 virtual processors. Under **Security**, enable Secure Boot using the Microsoft Windows template and enable Trusted Platform Module.

Checkpoint. The VM should now have a bootable Windows ISO, 8 GB RAM, two virtual processors and an 80 GB virtual disk. The Default Switch provides connectivity; all three tests will use local accounts inside the VM.

Microsoft: creating a VM in Hyper-V

Prepare Windows and accounts

Start the guest, complete its installation and create the two local accounts used by the tests.

- 1 Start the installer.** Right-click the VM in Hyper-V Manager and choose **Connect**, then **Start**. Press a key inside the VM if prompted to boot from the ISO. Follow Windows Setup and install onto the VM's virtual disk.
- 2 Complete initial setup.** Use English for the interface and create a local administrator named **LabAdmin**. If the work or school sign-in screen offers **Sign-in options > Domain join instead**, use it to create the local account. Setup choices vary by image version.
- 3 Confirm the administrator.** Once at the desktop, open `lusrmgr.msc` using **Win + R**. Open **Users > LabAdmin > Member Of** and confirm Administrators membership. Continue from the LabAdmin session.
- 4 Create the standard account.** In **Users**, right-click and choose **New User**. Enter **LabUser** and a non-empty password you can remember. Clear **User must change password at next logon**, then Create and Close.
- 5 Check account membership.** Open **LabUser > Member Of**. Keep it in **Users**; it should not be in Administrators. Confirm the account is enabled and is not locked out.

Account	Role in the lab
LabAdmin	Local administrator; the session requesting the tests
LabUser	Local standard user; target of the incorrect and correct password tests
Utentelnesistente	Deliberately absent; do not create this account

- 6 Take a checkpoint.** In Hyper-V Manager, right-click the VM and choose **Checkpoint**. Give it a clear name, such as `01-Windows-installed`.

Ready to proceed? You can sign in as LabAdmin, LabUser is a standard local account with a known password, and Utentelnesistente is absent. The guest computer name may differ from the original DESKTOP-N7PEBGK.

Enable auditing and run tests

Stay inside the VM as LabAdmin. Record the local time immediately after each attempt.

Enable Audit Logon

Press **Win + R**, enter `secpol.msc` and open **Advanced Audit Policy Configuration > System Audit Policies > Logon/Logoff > Audit Logon**. Select **Success** and **Failure**, then Apply and OK.

Open **Command Prompt** (`cmd`) from the LabAdmin desktop. The commands use Command Prompt syntax: `%COMPUTERNAME%` inserts the guest name. Run each command separately. Password prompts display no characters.

A. Enter an incorrect password

```
runas /user:%COMPUTERNAME%\LabUser cmd
```

At the password prompt, enter a deliberately incorrect password once. Expect failure and no new window. Write down the time. Stop here before running the next test.

B. Enter the correct password

```
runas /user:%COMPUTERNAME%\LabUser cmd
```

Run the command again in the original window and enter the correct LabUser password. In the **new command window**, run `whoami`. Confirm it returns your guest name followed by `\labuser`. Record the time, then close that new window.

C. Use a nonexistent username

```
runas /user:%COMPUTERNAME%\UtenteInesistente cmd
```

Back in the original LabAdmin window, run this command and enter any non-empty test password. Expect failure. Record the time. The original tests A and C both returned error **1326**; the logs distinguish their causes.

Keep the same method. Use plain `runas` as shown, without `/netonly` or `/savecred`. One deliberate incorrect-password attempt is enough. The original A-to-B gap was about 13 seconds; your interval does not need to match.

Find your events

Use the Security log on the guest. Background Windows activity will appear alongside your three tests.

- 1 Open Event Viewer.** Press **Win + R**, enter `eventvwr.msc`, and open **Windows Logs > Security**. Approve elevation as LabAdmin if prompted.
- 2 Filter the log.** Choose **Filter Current Log**. Enter 4624,4625 in the Event IDs field, select a time range covering your tests, and click OK. Refresh if the newest entries do not appear.
- 3 Find test A.** Inspect a 4625 near the time of your incorrect-password attempt. Double-click it and check **General > Account For Which Logon Failed > Account Name**. It should be LabUser.
- 4 Find tests B and C.** Locate a later 4624 with **New Logon > Account Name = LabUser**, then a 4625 for **Utentelnesistente** at the third test time. Keep the three events distinct.
- 5 Record identifiers.** Open **Details > XML View** to read EventRecordID and TimeCreated. General shows local time; XML timestamps ending in Z use UTC.

Test	Event to locate	Target account
A - Incorrect password	4625 / Audit Failure	LabUser
B - Correct password	4624 / Audit Success	LabUser
C - Unknown username	4625 / Audit Failure	Utentelnesistente

Use your own evidence. Record your actual times, computer name, Record IDs and session IDs. They will differ from the original screenshots. Additional events are normal; match the target account and time instead of selecting the first event in the list.

Use the next three pages as visual checkpoints. The screenshots are the original lab captures with crops and thin blue underlines; their log text has been preserved. Account names and filenames remain exactly as recorded.

Check the incorrect password

Test A: open the selected 4625 event and compare the highlighted fields with your own result.

Where to look in General	Expected result
Account For Which Logon Failed	Account Name: LabUser
Failure Information	Status: 0xC000006D; Sub Status: 0xC000006A
Logon Type / Network Information	Original result: Type 2; source ::1

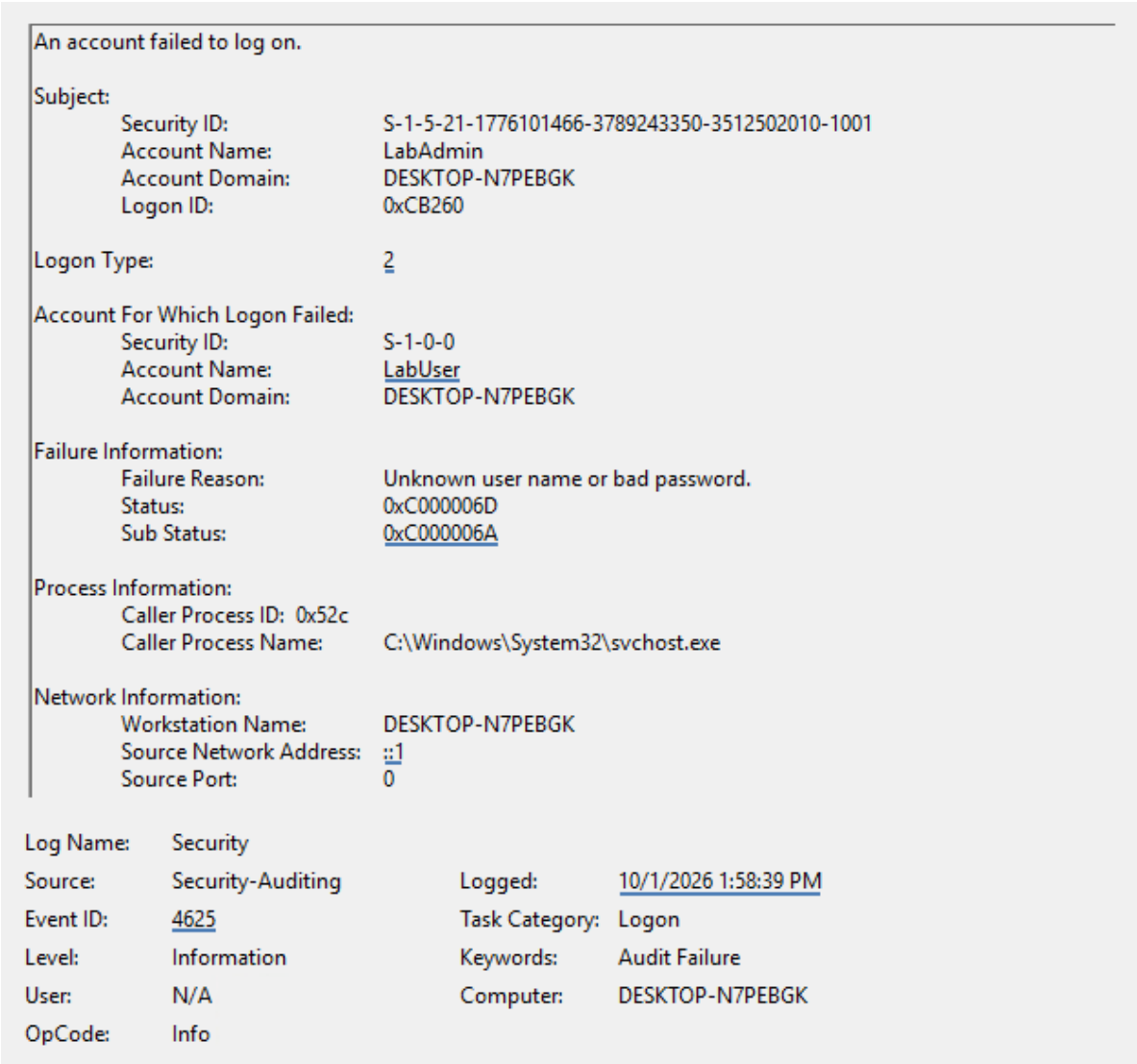


Figure 1. Original test A: Event 4625 at 13:58:39, 1 October 2026. The substatus identifies the incorrect-password failure.

The generic failure message alone is not enough. Verify the target account and Sub Status before moving on. Subject identifies the requesting context; it is not the target account.

Check the successful logon

Test B: open the later 4624 event for LabUser. Confirm that it corresponds to the correct-password attempt.

Where to look in General	Expected result
New Logon	Account Name: LabUser
Logon Information	Original result: Logon Type 2; Elevated Token No
Network Information	Original source address: ::1

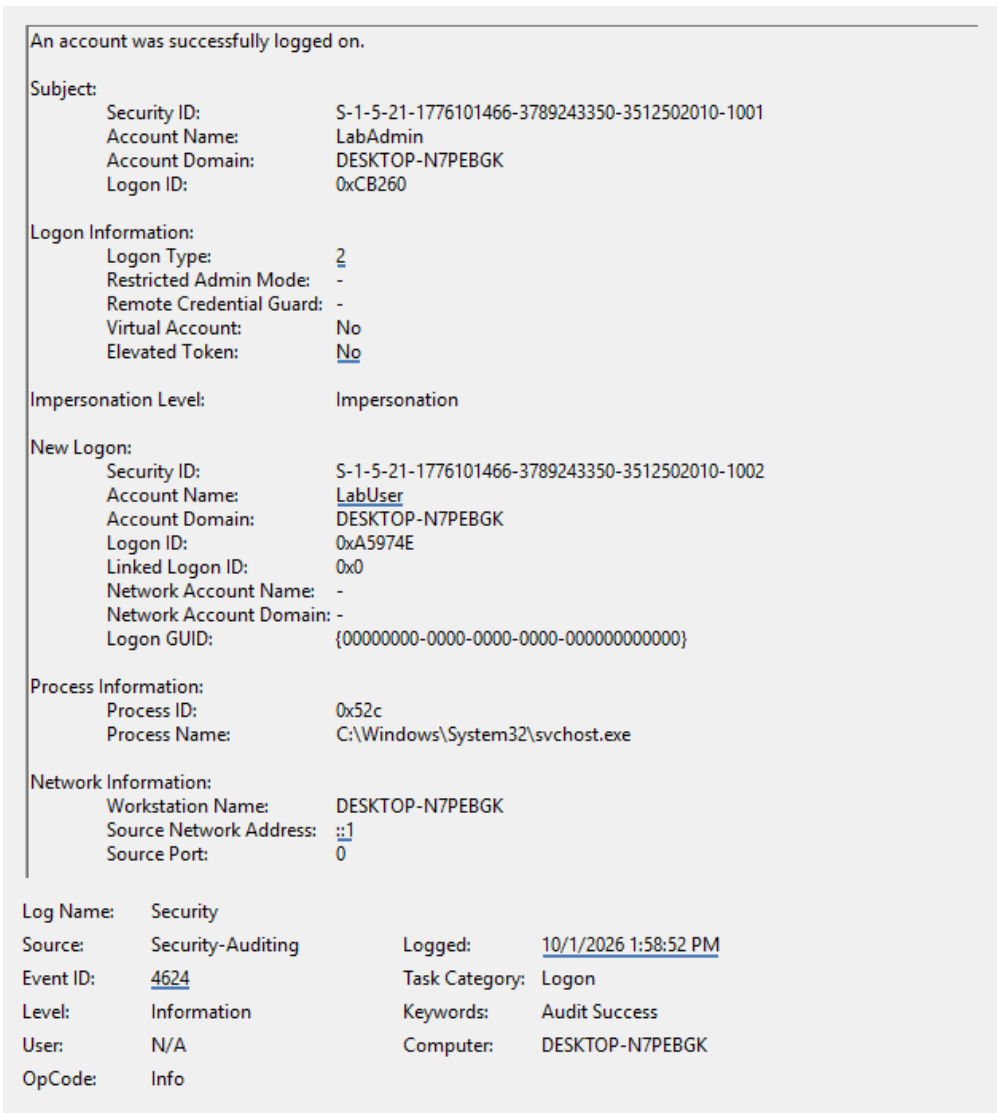


Figure 2. Original test B: Event 4624 at 13:58:52. It follows the failed LabUser attempt by about 13 seconds.

Compare the account, times, requesting session and source with test A. Your new LabUser Logon ID will be different from the original value. No Status or Sub Status failure code is expected in this successful event.

Check the unknown username

Test C: open the 4625 generated by the attempt using UtentInesistente, a name that was never created.

Where to look in General	Expected result
Account For Which Logon Failed	Account Name: UtentInesistente
Failure Information	Status: 0xC000006D; Sub Status: 0xC0000064
Logon Type / Network Information	Original result: Type 2; source ::1

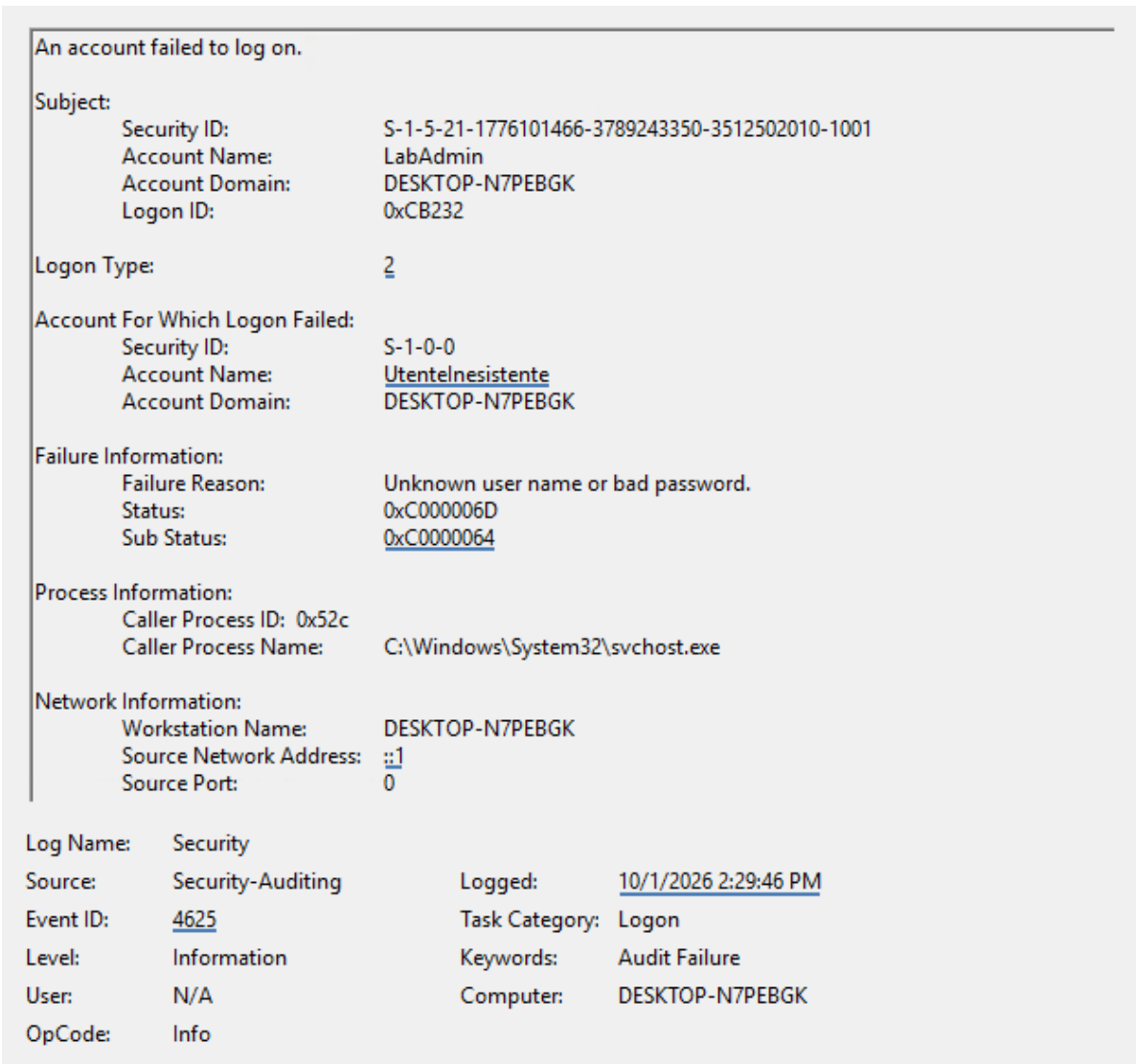


Figure 3. Original test C: Event 4625 at 14:29:46. Sub Status differs from test A even though the generic message is the same.

You should now have two failed attempts with different substatus codes and one successful attempt. If another code appears, check the actual account state and command rather than changing the event to fit the example.

Export evidence and finish

Keep the selected events and record the results of your own run before shutting down the VM.

- 1 **Create an evidence folder.** On the guest desktop, create **SOC-Lab-Evidence**. Select one event at a time in Event Viewer and choose **Save Selected Events**. Use EVTX format and the corresponding filename below.

Test	Evidence filename used in the project
A	01-accesso-fallito.evtx
B	02-accesso-riuscito.evtx
C	03-utente-inesistente.evtx

- 2 **Verify the exports.** If prompted, include display information in English. Keep any **LocaleMetaData** folder alongside the exports. Use **Open Saved Log** to confirm that each saved EVTX contains the intended event.
- 3 **Save screenshots and a timeline.** Capture the target, Event ID, time, logon type and relevant failure codes. Record the real times and Record IDs from your run. Keep the exported events with your lab evidence; the original project repository contains screenshots and documentation.

Completion checklist

A: 4625 for LabUser, Sub Status 0xC000006A. **B:** 4624 for LabUser and the new window confirmed by whoami. **C:** 4625 for UtenteInesistente, Sub Status 0xC0000064. All three selected events have been saved and reopened.

If something fails	First checks
No matching events	Guest Security log; time filter; Refresh; Audit Logon Success and Failure
Correct password rejected	Local username; password; account enabled/unlocked; no forced password change
runas cannot start	Read its error; inspect Secondary Logon in services.msc

Documentation

[Microsoft: Audit Logon](#)

[Microsoft: event 4624](#)

[Microsoft: event 4625 and failure codes](#)

[Microsoft: runas](#)

[Project: analysis report and annotated screenshots](#)